Recent Advances in Plasmonics and Nanophotonic Realizations

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Abstract

Plasmonics has undergone a radical transformation over the past decade, evolving from the study of passive metallic nanostructures supporting surface plasmon polaritons (SPPs) and localized surface plasmon resonances (LSPR) into a broad, multidisciplinary field that underpins quantum information science, solar energy conversion, advanced sensing, and next-generation nanofabrication. This review synthesizes the most significant developments across eight interconnected frontiers: (i) the paradigmatic shift from passive to active and reconfigurable plasmonics driven by epsilon-near-zero (ENZ) physics and time-varying media; (ii) the emergence of alternative plasmonic materials—transition metal nitrides, transparent conducting oxides, and MXenes—that transcend the limitations of noble metals; (iii) quantum plasmonics and on-chip single-photon sources; (iv) solar energy conversion via plasmonic photocatalysis and interfacial solar-steam generation; (v) wearable surface-enhanced Raman spectroscopy (SERS) and point-of-care diagnostics; (vi) thermal management and loss-mitigation strategies for plasmonic integrated circuits; (vii) active tunable metasurfaces employing phase-change materials and hybrid Mie-plasmonic resonators; and (viii) scalable nanofabrication through roll-to-roll nanoimprint lithography and artificial-intelligence-assisted design. For each frontier, we identify the key breakthroughs, benchmark performance metrics, and outstanding challenges, providing a roadmap for the translation of plasmonic research from the laboratory to practical devices.

Keywords: Surface plasmon polaritons · Epsilon-near-zero materials · MXenes · Quantum plasmonics · Active metasurfaces · Solar energy conversion · SERS · Nanoimprint lithography

1 Introduction: The Paradigmatic Shift from Passive to Active Plasmonics

The discipline of plasmonics has traditionally been defined by the study of surface plasmon polaritons (SPPs) and localized surface plasmon resonances (LSPR), phenomena arising from the collective oscillation of free electrons in metallic nanostructures coupled to electromagnetic fields [1, 2]. For decades, the field was dominated by noble metals—primarily gold and silver—owing to their high quality factors and resonant behaviour within the visible and near-infrared (NIR) spectra. However, the contemporary landscape of plasmonic research has undergone a radical transformation.

A clear transition is now evident from purely passive components to *active* and *reconfigurable* architectures whose optical response can be modulated post-fabrication through external stimuli such as electrical bias, thermal cycles, mechanical strain, or phase transitions [3, 4]. This evolution is fundamentally driven by the necessity to integrate plasmonic functionalities into practical electronic and photonic circuits. While conventional metallic nanostructures offer extraordinary light confinement, their fixed geometries limit utility in dynamic applications such as optical switching, adaptive lenses, and reconfigurable sensors [3, 4].

Modern theoretical frameworks now incorporate the *Active Metamaterials Roadmap*, which highlights a trajectory towards “timetronics” and “time crystals”—where material properties are modulated not merely in space but also in time—enabling non-reciprocal light propagation and the suppression of dissipation through the quantum Zeno effect [5, 6].

1.1 The Drude Model and Dynamic Permittivity Engineering

The physics of the active-plasmonics shift rests on manipulation of the dielectric function. The classical Drude model describes the metallic response as

$$\varepsilon(\omega) = 1 - \frac{\omega_{\mathrm{p}}^2}{\omega^2 + i\gamma\omega}\,, \tag{1}$$

where ω_{p} is the plasma frequency and γ is the damping constant. Recent advances allow dynamic tuning of ω_{p} or γ through carrier injection or structural alteration of the lattice [2, 7]. This capability is critical for achieving *epsilon-near-zero* (ENZ) states, where the real part of the permittivity vanishes, leading to unprecedented nonlinearities and the ability to control the phase of light over deep-subwavelength distances [8, 9].

The ENZ regime is particularly compelling for active modulation. When $\mathrm{Re}[\varepsilon] \approx 0$, the phase velocity of light diverges, enabling uniform phase accumulation across an entire thin film regardless of its physical geometry. This phenomenon, exploited in transparent conducting oxide (TCO) waveguide modulators, allows intensity and phase switching with extinction ratios exceeding 45 dB/μm at voltages as low as $\sim$1 V [8, 10].

2 Advanced Material Systems Beyond Noble Metals

The dominance of gold and silver is being challenged by a new generation of "designer" plasmonic materials. Although noble metals possess high conductivity, they suffer from significant Ohmic losses, poor CMOS compatibility, and limited spectral tunability in the infrared (IR) range [1, 11]. The current research frontier is defined by transition metal nitrides (TMNs), transparent conducting oxides (TCOs), and two-dimensional (2D) materials such as MXenes [11–13].

Table 1: Summary of plasmonic material systems and their key properties. Q = quality factor; CMOS = complementary metal-oxide-semiconductor; NIR = near infrared; Mid-IR = mid infrared.

Material	**Spectral Range**	**Key Advantage**	**Integration**
Au, Ag	Visible	Highest Q factors	Poor (diffusion)
TiN, ZrN	Vis–NIR	Refractory, high thermal stability	High (CMOS)
AZO, ITO	NIR	Electrical tunability (ENZ)	High (foundry)
$Ti_3C_2T_x$	NIR–Mid-IR	Hydrophilic, 2D morphology	High (solution)
Kagome FeSn	Mid-IR	Dirac-band conductivity	Emerging

2.1 Transition Metal Nitrides and Transparent Conducting Oxides

Transition metal nitrides (TMNs) such as titanium nitride (TiN) and zirconium nitride (ZrN) have emerged as *refractory* plasmonic materials [14]. These compounds offer high thermal stability and compatibility with standard semiconductor manufacturing, making them ideal for high-power applications and integration into silicon photonic chips [14]. Unlike gold, which melts at ∼1065 °C, TiN retains its plasmonic properties well above 2000 °C, opening the door to thermally robust nanophotonic devices [7, 14].

In the NIR, transparent conducting oxides such as aluminium-doped zinc oxide (AZO) and indium tin oxide (ITO) provide a unique platform for electrical tunability [8, 10]. Hybrid plasmonic waveguides employing a thin AZO layer as the active medium exploit the ENZ effect: a low voltage (∼1 V) modulates the carrier density in the AZO layer to transiently reach the ENZ condition, facilitating phase and intensity modulation with high extinction ratios [10]. This mechanism outperforms traditional thermo-optic or electro-optic effects because it directly modulates the plasmonic resonance rather than inducing subtle refractive index changes in a bulk dielectric [8].

2.2 The Rise of MXenes in Sustainable Nanotechnology

MXenes—a diverse family of 2D transition metal carbides and nitrides—have been identified as "rising stars" in sustainable nanotechnology [13]. Discovered in 2011, MXenes such as $Ti_3C_2T_x$ exhibit high metallic conductivity, high breakdown voltages, and a hydrophilic nature that distinguishes them from other 2D materials such as graphene or transition metal dichalcogenides (TMDs) [12, 15]. Their aqueous processability enables fabrication of plasmonic structures via 3D printing and roll-to-roll manufacturing at room temperature [13, 15].

The optical properties of MXenes are highly tunable via surface termination chemistry. Single-layer Ti_3C_2 and Nb_4C_3 show strong absorption in the infrared [15], which has been exploited for photothermal cancer therapy and solar-driven desalination [16]. Recent studies have successfully integrated MXenes as alternative plasmonic coatings in one-dimensional photonic crystal platforms to support Tamm plasmon polaritons (TPP) [17], demonstrating high optical asymmetry and large high-to-low reflection ratios with minimal dielectric layers.

3 Quantum Plasmonics and On-Chip Single-Photon Generation

The intersection of plasmonics and quantum optics—*quantum plasmonics*—exploits deep-subwavelength confinement to enhance quantum light-matter interactions. This field underpins scalable quantum communication networks and photonic quantum computing [18, 19].

3.1 Deterministic Single-Photon Sources

A critical requirement for quantum systems is a reliable *single-photon emitter* (SPE) that generates precisely one photon per excitation cycle with high purity and indistinguishability [19]. While traditional methods were probabilistic—such as spontaneous parametric down-conversion (SPDC)—recent breakthroughs focus on deterministic sources using semiconductor quantum dots (QDs) [18, 19].

Table 2: Benchmark comparison of on-demand single-photon source technologies. QD = quantum dot; SPDC = spontaneous parametric down-conversion; PIC = photonic integrated circuit; MIR = mid-infrared; QKD = quantum key distribution.

Technology	**Mechanism**	**Temperature**
QD Circular Bragg	Shaped laser pulse	Cryogenic
Rare-Earth Ion Fibre	Nd^{3+} excitation	Room temp.
Phonon-Assisted	Heralded	Intermediate
SPDC Metasurface	$\chi^{(2)}$ nonlinear	Room temp.

Advances in nanophotonic engineering have redefined the performance limits of single-photon sources. Integration with circular Bragg resonators (CBRs) and photonic crystal (PC) cavities enhances photon extraction efficiency while maintaining high-fidelity polarisation entanglement [18, 19]. A recent landmark demonstration produced a chip that reliably emits single photons on demand by coupling quantum dots to carefully shaped laser pulses [20]. This advance is pivotal for quantum key distribution (QKD), as it eliminates the multi-photon "replication" risk inherent in probabilistic sources [20]. Simultaneously, researchers have demonstrated a room-temperature scheme for generating single photons from individual rare-earth ions in optical environments, offering low-loss insertion into existing telecommunications networks [21].

3.2 Entanglement Mechanics and High-Fidelity Quantum State Tomography

Bipartite entanglement in nanophotonic systems is achieved through the Hong-Ou-Mandel (HOM) effect, where two SPPs interfere on a plasmonic beamsplitter [22]. Quantum state tomography of these systems has confirmed that plasmonic platforms can sustain quantum interference despite their inherently dissipative nature [22]. The preservation of energy-time entanglement in plasmon-assisted light transmission has been verified experimentally, with measured concurrence values proving plasmonic coherence across multiple scattering events [23].

The shift from the traditional biexciton-exciton (XX-X) cascade to spontaneous two-photon emission (STPE) represents a major conceptual advance [19]. In STPE, a biexciton decays directly to the ground state, emitting a photon pair simultaneously. This suppresses multi-pair emission and achieves high brightness per excitation pulse, circumventing limitations arising from solid-state dephasing in the cascade model [18, 19].

4 Solar Energy Conversion and Environmental Remediation

Plasmonic nanomaterials have become indispensable in solar energy harvesting owing to their LSPR-driven broadband sunlight absorption and efficient photothermal conversion [24].

4.1 Enhanced Photocatalytic Hydrogen Production

Green hydrogen production through photocatalytic water splitting is a flagship application of plasmonic composite systems. By coupling plasmonic nanoparticles (e.g., Au, Pd) with wide-bandgap semiconductors such as TiO_2 and ZnO, researchers have surmounted the thermodynamic barriers of traditional photocatalysis [24, 25]. Hot-electron injection from the metal nanoparticle into the conduction band of the semiconductor drives redox reactions under sub-bandgap illumination, dramatically extending the spectral utilisation window [25].

A notable breakthrough involves *defect repair* in ZnO heterojunctions, where individual gold atoms are employed to fill oxygen vacancy defects [25]. This passivation increases donor charge density and extends photo-electron lifetimes, resulting in substantially enhanced hydrogen output rates compared to bare ZnO. The strategy highlights the importance of atomic-scale engineering in plasmonic catalysis.

4.2 Solar-Driven Interfacial Evaporation and Desalination

Interfacial evaporators exploiting plasmonic materials have achieved conversion efficiencies and evaporation rates that were previously considered unattainable [26]. Table 3 benchmarks representative systems.

Table 3: Performance of plasmonic interfacial evaporator systems under 1-sun illumination (1 kW/m^2).

System	Efficiency (%)	Rate ($kg\,m^{-2}h^{-1}$)
Al NP 3D Assembly	$\sim$90	$\sim$1.0
TiN Nanocomposites	$\leq$99	1.76
Pd/Semiconductor	>89	>1.5
MXene/Au@$Cu_{2-x}S$	96.1	2.023

The three-dimensional aluminium nanoparticle assembly reported by Zhou et al. [26] pioneered the concept of plasmon-driven interfacial steam generation, achieving broadband solar absorption through coupled dipolar resonances across the visible spectrum. Subsequent work combining multiple plasmonic media (MXene, gold, and $Cu_{2-x}S$) has pushed light-to-heat conversion efficiencies above 96%, with evaporation rates of $\sim$2 $kg\,m^{-2}\,h^{-1}$, demonstrating the synergy achievable through plasmonic heterostructures [15].

5 Wearable Sensors and Point-of-Care Diagnostics

Plasmonic sensing—in particular surface-enhanced Raman spectroscopy (SERS)—has transitioned from specialised laboratory equipment to wearable healthcare solutions, leveraging nanoscale light-matter interactions to detect electrolytes, metabolites, hormones, and proteins in sweat or interstitial fluid [1, 27].

5.1 Integrated SERS and Wearable Platforms

Wearable plasmonic sensors are being embedded into smart contact lenses for continuous monitoring of intraocular pressure and glucose levels [28]. These platforms combine material innovations with machine-learning algorithms to interpret complex spectral data, improving specificity and reducing false-positive rates [28].

SERS substrates can be engineered to provide electromagnetic field-enhancement factors exceeding 10^8, enabling single-molecule sensitivity [27]. A comprehensive understanding of these enhancement factors, spanning both chemical and electromagnetic contributions, is essential for the design of quantitative wearable sensors [27]. Systematic characterisation of the axial SERS response reveals a Lorentzian signal profile along the optical axis, underlining the importance of focus precision for reproducible wearable measurements [29].

5.2 MXene-Based Biosensing

MXenes have emerged as superior substrates for electrochemical biosensors due to their large surface area, high electrical conductivity, and abundant surface functional groups that facilitate target capture [13]. The excellent electrochemical performance of $Ti_3C_2T_x$ was established through cation intercalation studies demonstrating high volumetric capacitance [16], properties equally valuable in sensing applications.

Antibody-functionalised $Ti_3C_2T_x$ nanosheets have been deployed for point-of-care detection of biomarkers at clinically relevant concentrations critical for screening of nutrient deficiencies and toxicity in resource-limited healthcare settings [15, 28]. The combination of MXene substrates with plasmonic enhancement further amplifies signal-to-noise ratios, promising multiplex sensing of biomarkers from a single micro-volume sample [13].

6 Thermal Management and Loss Mitigation in Integrated Circuits

As semiconductor devices approach kilowatt-class thermal design power—particularly in AI accelerators and three-dimensional integrated circuits (3D ICs)—thermal management has become the primary bottleneck for system performance [30, 31]. Plasmonic integrated circuits, operating at high speeds and subject to Ohmic losses, are especially susceptible [2, 32].

6.1 Cooling Inside the Silicon and Microfluidic Approaches

The industry is shifting toward "cooling inside the silicon," where microchannels are etched directly into silicon substrates to act as a "circulatory system" for coolant [30]. This approach routes coolant to within micrometres of active transistors or plasmonic modulators, often utilising two-phase (liquid-to-vapour) cooling to dramatically reduce thermal resistance [30].

Table 4: Recent thermal management advances relevant to plasmonic integrated circuits. TIM = thermal interface material; BAs = boron arsenide.

Technology	Key Metric	Status
Microfluidic direct-to-chip	Kilowatt-class extraction	Emerging
Graphene TIM	$>10\times$ greases	Active
Boron Arsenide	$1300\ \mathrm{W\,m^{-1}K^{-1}}$	R&D
Solid-State Transistor	Tunable heat flow	Experimental

Materials with ultrahigh thermal conductivity, such as boron arsenide (BAs, $1300\ \mathrm{W\,m^{-1}K^{-1}}$) and boron phosphide, have set new conductivity benchmarks [33]. Phonon band-structure engineering has reduced thermal boundary resistance (TBR) at GaN interfaces by more than eightfold compared to GaN/diamond interfaces, markedly lowering hotspot temperatures in high-power devices [33]. Graphene-based thermal interface materials provide more than a tenfold improvement over conventional thermal greases, accelerating heat spreading in densely integrated photonic chips [31].

6.2 Compensating Losses via Optical Gain Integration

To address the intrinsic Ohmic losses of plasmonic systems, a powerful strategy employs optical gain media in the near field of plasmonic structures. De Leon and Berini demonstrated amplification of long-range SPPs by embedding the plasmonic waveguide in a dipolar gain medium, achieving net signal amplification that compensates propagation losses [34]. This constitutes a viable pathway for loss-sensitive applications in photonic circuits and sub-diffraction-limit imaging [34]. Combined with recent proposals for synthetic optical excitation at complex frequencies, these techniques enable dramatically enhanced effective propagation distances without physical gain saturation [32].

7 Active Tunable Metasurfaces and Reconfigurable Optics

Real-time optical reconfigurability defines the next generation of multifunctional optoelectronic devices. Active metasurfaces employ electrical modulation, mechanical strain, and thermal phase transitions to dynamically tailor amplitude, phase, and polarisation of light [3, 4, 35].

7.1 Vanadium Dioxide and Phase-Change Chalcogenides

Vanadium dioxide (VO_2) is a volatile phase-change material (PCM) that undergoes a metal-insulator transition at 68 °C, switching from a monoclinic (insulating) to a rutile (metallic) phase [4, 36]. Kats et al. demonstrated that a sub-wavelength VO_2 film can provide thermally switchable, near-unity absorption across the mid-infrared, establishing a benchmark for ultra-thin tunable perfect absorbers [36]. Upon thermally inducing the phase transition, absorption resonances can be spectrally tuned across tens of nanometres [36].

Non-volatile PCMs such as GeSbTe (GST) and GeSbSeTe (GSST) retain their phase state after the initial stimulus is removed, making them ideal for programmable photonics and optical memory [35]. Wang et al. demonstrated optically reconfigurable metasurfaces and photonic devices using GST phase transitions, achieving on-demand pixel-wise control of amplitude and phase, enabling varifocal lensing and dynamic holography [3, 35].

7.2 Hybrid Dielectric-Plasmonic Resonators

Dielectric nanoresonators are preferred over metallic counterparts for their lower energy dissipation, but suffer from less confined resonances and lower absolute field enhancement [37]. Hybrid Mie-plasmonic (MP) systems resolve this trade-off by combining the low losses of Mie-resonant dielectrics with the strong field confinement of plasmonics [37].

Supercavity modes with record-high quality factors arise from the coupling of Mie resonances with Fabry-Pérot-plasmonic modes near higher-order anapole conditions—a prediction confirmed experimentally by Rybin et al. with $Q > 260$ in silicon nanodiscs [38]. These high-Q resonances dramatically enhance nonlinear optical processes: in coupled dielectric-semiconductor nanoresonators, second-harmonic generation (SHG) signal intensities have been boosted by over an order of magnitude, a milestone for miniaturised frequency converters and entangled photon pair sources [39].

8 Scalable Nanofabrication and Industrial Adoption

The translation of plasmonic research from laboratory to manufacturing is enabled by scalable techniques such as nanoimprint lithography (NIL) and roll-to-roll (R2R) processing [40, 41].

8.1 Roll-to-Roll Nanoimprint Lithography

R2R-NIL provides continuous patterning of complex nanostructures on flexible polymers, glass foils, and thin films [41, 42]. UV-curing imprint engines can replicate gratings, metasurfaces, and diffractive optical elements with sub-100 nm resolution at linear speeds from 0.005 to 50 m/min [42]. This technology is currently being adopted for augmented and virtual reality (AR/VR) waveguide manufacturing, which demands high refractive indices, low surface roughness, and precise slanted or blazed grating geometries [41].

Table 5: Comparison of patterning technologies for plasmonic nanostructure manufacturing. EBL = electron-beam lithography.

Metric	R2R-NIL	R2P-NIL	EBL
Speed (m/min)	0.005–50	Medium	Slow
Throughput (m^2/yr)	1.5×10^4–4×10^6	Medium	Low
Resolution	<100 nm	<100 nm	<10 nm
Substrate	Flexible	Rigid	Wafer

Template stripping—a technique reported by Nagpal et al. for producing ultrasmooth patterned metal films [43]—has been extended to stretchable polydimethylsiloxane (PDMS) substrates, creating gold nanohole arrays and pyramid-tip antennas whose optical resonances can be mechanically tuned by stretching the underlying film [43]. This mechanical tunability adds a new degree of freedom unavailable to rigid plasmonic substrates.

8.2 Additive Manufacturing and AI-Enhanced Design

The integration of plasmonic materials—particularly MXenes—into 3D printing workflows enables fabrication of intricate architectures for energy storage, electromagnetic shielding, and electronics [13, 15]. Artificial intelligence (AI) and machine-learning (ML) algorithms are now routinely employed to design and optimise plasmonic metasurfaces, predicting transmission and phase responses with low mean-squared errors across the visible and NIR spectral ranges [44]. Inverse design via deep neural networks and generative adversarial models dramatically reduces the design cycle time from weeks to minutes [44], democratising high-performance metasurface development beyond specialist computational electromagnetics groups.

9 Outlook and Conclusions

Plasmonics stands at a pivotal juncture. The field has matured from fundamental studies of light-metal interactions into a broad technology platform that bridges quantum science, energy, healthcare, and semiconductor manufacturing. Several overarching trends will define the next decade:

1. **Time-varying and active media.** The integration of ENZ materials and phase-change media with time-varying electromagnetic environments will unlock nonreciprocal optical devices, photonic time crystals, and Floquet-engineered metamaterials [6, 9].

2. **Sustainable, CMOS-compatible materials.** The replacement of noble metals with TMNs, TCOs, and MXenes will enable mass production of plasmonic components within existing semiconductor foundry infrastructure [11, 14].

3. **Quantum advantage at room temperature.** Room-temperature single-photon emission from rare-earth ions [21], combined with plasmonic enhancement of light-matter coupling, will accelerate the deployment of quantum-secure communications.

4. **Convergence with AI.** Inverse design algorithms trained on large experimental datasets will enable the automated discovery of new plasmonic geometries with functionalities inaccessible through intuition-guided design [44].

5. **Wearable and implantable photonics.** The mechanical flexibility of MXene and NIL-patterned substrates, combined with on-chip signal processing, will realise truly wearable plasmonic spectrometers for continuous personal health monitoring [13, 28].

6. **Thermal co-design.** As photonic integration densities rival those of microelectronics, thermal management will become a first-class design variable, necessitating co-simulation of optical, electrical, and thermal domains [30, 33].

In summary, the advances reviewed here demonstrate that plasmonics has transcended its origins as a curiosity of condensed matter physics to become an enabling platform technology with transformative impact across science and engineering. The challenges that remain—particularly loss mitigation, scalable quantum-coherent integration, and reliable nanofabrication at wafer scale—are formidable but tractable given the rapid pace of innovation documented in this review.

Declarations

Conflict of Interest The author declares no conflict of interest.

Data Availability Not applicable.

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